

FIN ST1272–ST1361

Frozen OA Dual-Dynamics Discrimination and Composite Robustness

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Six research rounds — 90 programmes
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This report freezes one explicitly conditional Operational/State package and turns the mathematical duality e^{-tA} versus e^{-itA} into an executable model-discrimination specification. All finite-count guarantees refer to the declared ideal hypotheses. Synthetic fixtures are not experimental evidence.

Abstract

Release 10.97 proved that FIN requires an independent operational/state package OA in addition to conversion and sector packages. ST1272–ST1361 freeze one minimal comparison: prepare vertex 0, evolve with the strict generator A , and measure return to vertex 0. The classical hypothesis uses the Markov semigroup e^{-tA} ; the quantum hypothesis uses the unitary group e^{-itA} .

The return probabilities have exact spectral forms

$$C(t) = \frac{1}{12} \sum_k e^{-\lambda_k t}, \quad Q(t) = \left| \frac{1}{12} \sum_k e^{-i\lambda_k t} \right|^2.$$

A million-point scan of $0 \leq t \leq 10$ together with analytic derivatives finds the strongest first separation at $t_* \approx 0.59453079$, where $|Q - C| = 0.4112409998$. This optimization is strong numerical evidence, not an interval-certified global uniqueness theorem. The executable protocol freezes $t = 0.6$, losing only approximately 2.3×10^{-5} of the separation.

Under perfect iid preparation and readout, the return event is Bernoulli with $p_C = 0.4123573972$ and $p_Q = 0.8235756226$. An integer likelihood-ratio rule using 29 clicks and choosing quantum for at least 19 returns gives both errors below one percent. Full twelve-vertex readout has Chernoff information 0.1364798000, exceeding the binary-return value 0.1002859998.

The ideal result does not survive arbitrary model broadening. Energy-basis dephasing supplies a scalar rate that exactly matches the classical return at one time; hence a single return probability cannot distinguish classical heat from the entire open-quantum family. A frozen four-time design (0.3, 0.6, 1.2, 2.0) remains separated in a numerical composite fit when the clock scale is independently constrained to $[0.9, 1.1]$. If the clock scale is allowed to range over $[0.1, 10]$, the open quantum family nearly mimics the classical curve. Clock calibration is therefore a necessary input, not an optional convenience.

The deliverable includes a hashed JSON protocol, a fail-closed validator, no-click accounting, synthetic positive/negative fixtures and a one-shot analysis rule. It does not include apparatus, raw natural events, independent custody or a laboratory claim. The result makes dual dynamics falsifiable in principle under a frozen OA while sharply defining the nuisance-model and evidence boundaries.

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1 Scope and epistemic hierarchy

This cycle does not attempt to decide the ontology from A . Release 10.96 proved that impossible on the A -only datum. It instead chooses two explicitly conditional hypotheses inside OA and asks whether their predictions differ.

Four levels are kept separate:

1. exact finite-model identities;
2. strong numerical optimization or composite fitting;
3. synthetic protocol/validator replay;
4. laboratory evidence, which is absent.

2 Round I: ST1272–ST1286 — frozen operational hypotheses

The common preparation is the selected vertex 0. The common coarse observable asks whether the final vertex equals 0.

Classical hypothesis H_C .

$$p(0) = \delta_0, qquad p(t) = e^{-tA}\delta_0, qquad C(t) = (e^{-tA})_{00}.$$

Quantum hypothesis H_Q .

$$\rho(0) = |0\rangle\langle 0|, qquad U_t = e^{-itA}, qquad Q(t) = |\langle 0|U_t|0\rangle|^2.$$

Both models use $t = t_{\text{phys}}/\tau_*$. The base hypotheses set dephasing and loss to zero and fix the clock factor to one. Those restrictions are part of the hypothesis, not consequences of A .

The full vertex distributions are

$$p_x^C(t) = (e^{-tA})_{x0}, qquad p_x^Q(t) = |(e^{-itA})_{x0}|^2.$$

They are normalized and nonnegative. Circulant symmetry makes all choices of the source vertex equivalent; this does not produce an absolute origin.

The protocol records a SHA-256 fingerprint of the rounded reconstructed matrix and freezes the optimization domain $0 \leq t \leq 10$ before any future data.

3 Round II: ST1287–ST1301 — spectral optimization

The strict spectrum reduces to seven values with multiplicities

$$1, 2, 2, 2, 2, 2, 1.$$

Consequently,

$$C(t) = \frac{1}{12} \left(1 + 2 \sum_{j=1}^5 e^{-\lambda_j t} + e^{-\lambda_6 t} \right)$$

decreases to $1/12$. The quantum amplitude is the analogous finite sum of phases, so $Q(t)$ is almost periodic rather than monotone.

For

$$D(t) = |Q(t) - C(t)|,$$

analytic differentiation gives, with $a(t) = 12^{-1} \sum_k e^{-i\lambda_k t}$,

$$Q'(t) = 2 \operatorname{Re}(\bar{a}a'), qquad Q''(t) = 2(|a'|^2 + \operatorname{Re}(\bar{a}a'')).$$

The numerical stationary point is

$$t_* \approx 0.5945307906, \text{quad}C(t_*) \approx 0.4152434795, \text{quad}Q(t_*) \approx 0.8264844793,$$

$$D(t_*) \approx 0.4112409998.$$

The second derivative of $Q - C$ is negative there. A grid of 1,000,001 points with step 10^{-5} finds the same maximum cell. A global derivative bound supplies a small cover gap, but floating evaluations are not an outward interval proof. The report therefore does not claim an exact unique global maximizer.

At the practical frozen time $t = 0.6$,

$$C = 0.4123573971501711, \text{quad}Q = 0.8235756226155311, \text{quad}D = 0.4112182254653600.$$

The loss relative to the candidate optimum is about 2.28×10^{-5} .

At t_* , every nonreturn quantum probability is below its classical counterpart, so the total-variation distance of the full vertex distributions equals the return gap numerically. Later, the sign can reverse because of unitary recurrence; three positive crossings occur in $(0, 10]$ near 5.6671, 7.0805 and 9.7075.

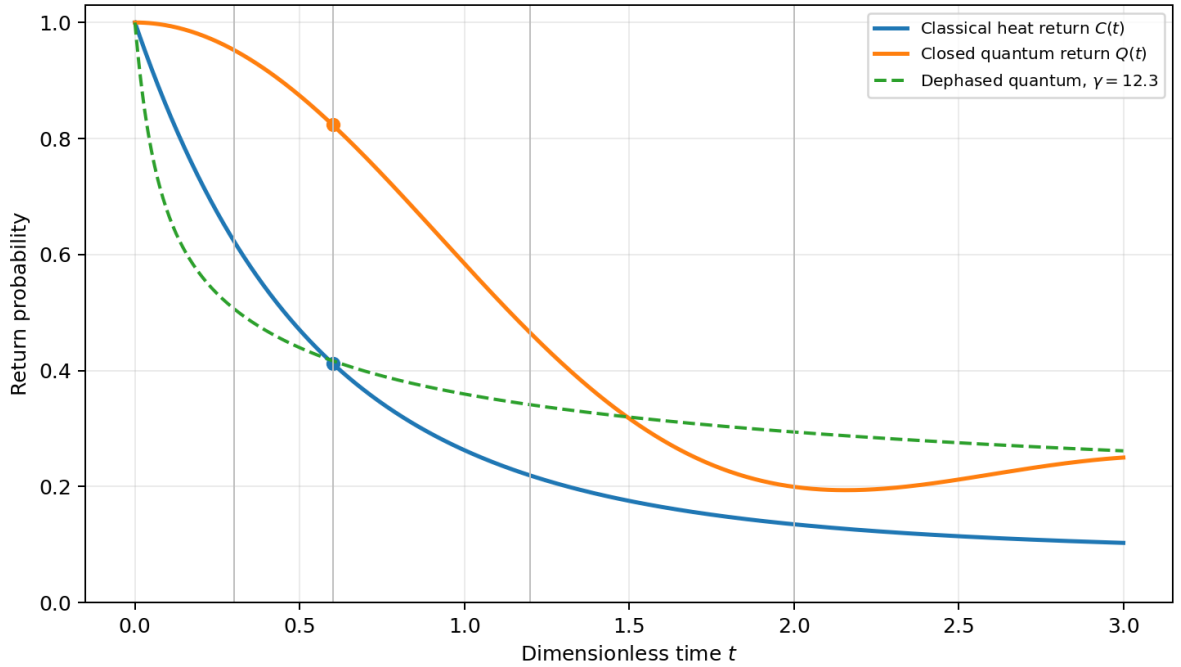


Figure 1: Frozen classical and closed-quantum return curves, with one strongly dephased quantum example. Vertical lines mark the four-time robust design.

4 Round III: ST1302–ST1316 — finite counts

At $t = 0.6$, repeated ideal return observations give

$$K \sim \text{Binomial}(n, p_C) \text{quad} \quad K \sim \text{Binomial}(n, p_Q).$$

The directional Kullback–Leibler divergences are

$$D(C\|Q) = 0.4218128334, \text{quad}D(Q\|C) = 0.3574422295,$$

and the Bernoulli Chernoff information is 0.1002859998. Since $p_Q > p_C$, the likelihood ratio is monotone in the return count.

Maximum target error	shots n	threshold k	$C \rightarrow Q$ error	$Q \rightarrow C$ error
5%	15	10	0.0420851	0.0354013
1%	29	19	0.0071823	0.0077975
0.1%	48	31	0.0009280	0.0008798

With 100 ideal shots, the best integer threshold gives maximum error below 4.9×10^{-6} . These are exact binomial-tail calculations in floating arithmetic, conditional on iid trials, exact preparation, exact time and a perfect effect.

Retaining the full vertex outcome increases Chernoff information to 0.1364798000. This is expected by data processing: binary return is a coarse-graining of the multinomial record.

5 Round IV: ST1317–ST1331 — nuisance falsification

5.1 Clock error

For the base hypotheses the return gap remains above approximately 0.408 through a $\pm 10\%$ window around $t = 0.6$. Thus moderate timing error alone does not threaten the ideal separation.

5.2 Dephasing no-go at one time

Consider energy-basis dephasing generated by A :

$$\rho_{jk}(t) = \rho_{jk}(0) \exp \left[-i(\lambda_j - \lambda_k)t - \frac{\gamma}{2}(\lambda_j - \lambda_k)^2 t \right].$$

For a suitable $\gamma \approx 12.6$, its return probability at the frozen time equals the classical return. Therefore:

Theorem 1 (single-time composite no-go). *One return probability at one time cannot distinguish strict classical heat from the family of unitary-plus-free-energy-dephasing quantum models.*

Strong dephasing does not become classical heat. Because of spectral degeneracies its limiting return is

$$\frac{1^2 + 5 \cdot 2^2 + 1^2}{12^2} = \frac{22}{144} = 0.152777\dots,$$

not $1/12$.

5.3 Four-time composite design

The robust design freezes

$$t \in \{0.3, 0.6, 1.2, 2.0\}, \text{ } qqquad 100 \text{ shots per time.}$$

Fitting dephasing $0 \leq \gamma \leq 100$ and a clock factor $0.9 \leq \alpha \leq 1.1$ gives a best numerical residual sum of squares 0.0523947 and maximum point residual 0.1506064. This is promising separation, but differential evolution is not a formal global minimax certificate.

Allowing $0.1 \leq \alpha \leq 10$ reduces the best maximum residual to roughly 0.0195. Thus an independently calibrated clock scale is indispensable.

Uniform detection loss disappears if records are conditioned only on clicks. The protocol must record attempts and no-clicks. In a toy uniform vertex- confusion model, mixing each distribution with the uniform distribution at rate ϵ shrinks the gap by $1 - \epsilon$.

6 Round V: ST1332–ST1346 — executable specification

The file `FIN_ST1332_ST1346_Protocol.json` freezes:

- the matrix fingerprint and hypotheses;
- the primary $t = 0.6$, $n = 29$, $k = 19$ rule;
- the four robust times and 100 shots per time;
- the clock and dephasing nuisance bounds;
- raw fields: timestamp, attempt, click/no-click, vertex, configuration and run;
- invalid-record rules and the one-shot analysis requirement.

The dependency-light validator rejects missing fields, inconsistent vector lengths, noninteger counts, returns exceeding clicks, clicks exceeding attempts, shot mismatch and missing frozen time. Invalid data produce `INVALID`, not a forced classical/quantum verdict.

Seeded synthetic classical and quantum fixtures pass validation and are classified correctly by the primary rule. A deliberately invalid fixture fails closed. These fixtures test code only; they are not observations.

Provider, registrar and analyst must be distinct; the protocol hash must be frozen before unblinding. Local code cannot supply staffed custody roles, independent holdout or natural events.

7 Round VI: ST1347–ST1361 — final verdict

Claim	Status	Boundary
spectral return formulas	[Proven]	finite frozen A
large separation near $t = 0.595$	[Strong numerical]	no interval global uniqueness
29-shot sub-one-percent rule	[Proven in ideal model]	iid/perfect OA assumptions
full vertex improves Chernoff information	[Proven/numerical]	frozen distributions
single-time open-quantum identification	[Refuted]	free energy dephasing mimics return
four-time calibrated composite separation	[Strong numerical]	nuisance box, not minimax-certified
executable protocol and validator	[Constructed/tested]	synthetic fixtures only
laboratory discrimination	[Absent]	no apparatus/raw events/custody

The frozen OA makes FIN’s dual dynamics operationally distinguishable in principle, but only inside a declared model library. It does not prove that either channel is nature’s dynamics.

The experiment concerns redistribution/mixing versus coherent unitary evolution inside the proposed total state. It does not test literal nonexistence or annihilation of the whole nadsoliton.

8 Recommended next programmes

1. interval-certify the time optimum and stationary-root inventory;
2. solve the exact composite minimax discrimination problem;
3. add state-preparation contamination bounds;
4. add a calibrated detector confusion matrix and drift model;

5. replace the binary statistic by the full multinomial likelihood;
6. derive valid sequential confidence/e-value procedures;
7. certify four-time identifiability over the complete nuisance box;
8. map every *OA* field to one candidate platform without claiming realization;
9. freeze an independent custody/hash/holdout transfer package;
10. obtain independent laboratory review before collecting data;
11. collect raw event-level records only after protocol freeze;
12. report failure against the predeclared alternative library without model repair.

Stop and return **INVALID** after a matrix-hash mismatch, clock outside the calibrated band, unrecorded no-clicks, invalid counts, post-unblinding model edits or unregistered nuisance mechanisms.

9 Final conclusion

The dual FIN dynamics is no longer merely a formal analogy after *OA* is frozen: it produces sharply different finite probabilities and an executable test. The ideal distinction is large. The scientifically more important falsification is that one time point cannot identify classical heat against a reasonable open-quantum family. Multiple times, calibrated clocks, no-click records and a frozen nuisance library are necessary. The resulting protocol is ready as a mathematical transfer specification, not as experimental evidence.

A Six-round ledger

Range	Object	Verdict
ST1272–ST1286	frozen classical/quantum <i>OA</i>	common preparation and return observable
ST1287–ST1301	spectral optimization	near-optimal $t = 0.6$, effect 0.4112
ST1302–ST1316	finite counts	29-shot ideal sub-one-percent rule
ST1317–ST1331	nuisance robustness	single-time dephasing no-go; four-time candidate
ST1332–ST1346	executable protocol	hash, validator, no-clicks and fixtures
ST1347–ST1361	synthesis	conditional readiness, no laboratory evidence

B Reproducibility

The authoritative result sets run from `FIN_ST1272_ST1286_Results.json` through `FIN_ST1347_ST1361_Results.json`. Each contains fifteen programmes and SHA-256 packet references. The combined test `test_fin_st1272_st1361.py` checks all ninety packets, statistical thresholds, nuisance status, protocol validation, figure production and nonpromotion gates.

C Nonclaims

No physical state category, SI clock, apparatus, raw event, independent custody, laboratory distinction, legacy bridge, selector, Standard Model, gravity, L_{total} or Theory of Everything is established. All future FIN PDFs remain English.

References

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